

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2023****Course: B. Tech.****Branch: E&TC Engineering****Semester: VI****Subject Code & Name: BTETPE604A_Y18 & CMOS Design****Max Marks: 60****Date: 23/1/2024****Duration: 2:00 to 5:00 PM****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.

	(CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Implement D flip-flop using static CMOS logic	1	6
B) Implement $Y = (A + B).(\bar{C} + \bar{D})$ function using 2 input NOR gates only & then draw the schematic using static CMOS design style. Assume inputs available only in true form.	2	6
C) Draw the schematic and layout for three input CMOS NOR gate, Use CMOS Lambda based design rules.	2	6
Q.2 Solve Any Two of the following.		12
A) Write a short note on CMOS scaling	1	6
B) Explain leakage power in CMOS inverter. What are ways to reduce it?	1	6
C) Implement the full adder using half adder.	2	6
Q. 3 Solve Any Two of the following.		12
A) What are various Short Channel Effects? How to avoid them?	1	6
B) Write electrical, logical effort, parasitic effort for four input NOR gate.	2	6
C) Explain CMOS λ -based design rules in detail.	1	6
Q.4 Solve Any Two of the following.		12
A) Implement T Flip-Flop using Static CMOS design style.	2	6
B) Explain CMOS Domino logic in detail. Compare it with static CMOS logic	2	6
C) Explain all the parasitic capacitances of the MOSFET in detail	1	6
Q. 5 Solve Any Two of the following.		12
A) Implement 2-bit magnitude comparator using static CMOS logic	1	6
B) Implement half adder combinational circuit using static CMOS logic.	2	6
C) Implement $\overline{AB + CD}$ gate using dynamic CMOS logic.	2	6

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